Pin pitch gauge: card header J4 and the Digilent HS3 plug

Print at 100 % (actual size). Do not use fit to page.

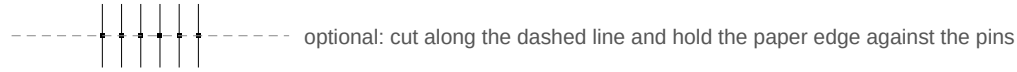
Step 1. Check the print scale: this bar must measure exactly 100 mm.

If it is not 100 mm the printer rescaled the page. Reprint before trusting the marks below.

Step 2. Card header J4 (six pins in a row). Lay the pins on each row. Only one row will match.

Row A: 2.54 mm pitch

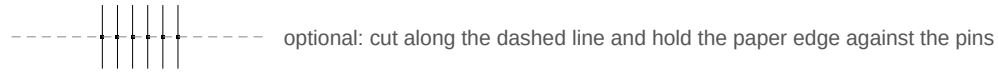
What the adapter is built for. All six pins should sit on the six black squares.



12.70 mm between the centres of the two end pins

Row B: 2.00 mm pitch

If the pins match this row instead, stop: the adapter's socket will not fit. Do not order.



10.00 mm between the centres of the two end pins

Step 3. HS3 plug (two rows of seven holes). Hold the end of the HS3 against each pattern.

Pattern C: 2.00 mm pitch

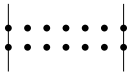
What the adapter is built for. All fourteen holes should sit on the fourteen dots.

12.00 mm between the first and last hole of a row, rows 2.00 mm apart

dashed box: outside of the adapter's header, 16.65 x 6.30 mm. The HS3 plug goes inside it.

Pattern D: 2.54 mm pitch

If the holes match this pattern instead, stop: this is not the Xilinx 2 mm plug. Do not order.



15.24 mm between the first and last hole of a row, rows 2.54 mm apart

Without a printer

Card header J4, centre of pin 1 to centre of pin 6	**12.70 mm = correct**	10.00 mm = wrong
Card header J4, outside faces of the two end pins (calipers)	**13.3 mm = correct**	10.5 mm = wrong
HS3 plug, centre of the first hole to centre of the seventh in a row	**12.00 mm = correct**	15.24 mm = wrong
HS3 plug, centre of one row to centre of the other	**2.00 mm = correct**	2.54 mm = wrong

Fit test for J4: a standard 0.1 inch jumper wire housing is 2.54 mm wide. Six side by side, or one 1x6 housing, should slide onto all six pins without spreading apart or bending a pin.

The pin at the 1V8 end of J4 is slightly bent on some cards. Straighten it gently before measuring.